[Sandia National Laboratories Aerial Photograph]

Place `cover_photo.jpg` in the same folder
and uncomment the includegraphics line above.

UC DAVIS

College of Engineering

Passive Temperature Compensation Device

SPONSORED BY: DR. BRADLEY BON, SANDIA NATIONAL LABORATORIES

PRELIMINARY MACHINE ARCHITECTURE REPORT

EME-185A/B

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Passive Temperature Compensation Device

By the

Noble Aggie Labs

Sponsored by Dr. Bradley Bon, Sandia National Laboratories

Preliminary Machine Architecture Report

EME-185A/B

Professor Jonathan Schofield

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Due 16 March 2026

Contents

1 Abstract **5**

2 System Architecture Selected **5**

2.1 Selection Criteria and Design Justification 6

2.2 Complete System Overview 6

2.3 Subsystem Overviews 8

2.3.1 Thermal Actuator 8

2.3.2 Valve 8

2.3.3 Housing and Fluid Interface 9

3 Discussion **9**

3.1 Design Requirement Updates 9

3.2 Requirements Rolldown to Subsystems 10

3.3 Subsystem Interactions and Technical Detail 11

3.4 Design Verification Calculations 11

3.4.1 Thermal Expansion and Actuator Displacement 11

3.4.2 Force Analysis Under Pressure Loading 12

3.4.3 Performance Relationship 12

3.4.4 Additional Analyses 12

4 Preliminary Bill of Materials **13**

5 Project Management Information **14**

5.1 Updated Task List 14

5.2 Gantt Chart 15

5.3 Critical Path Method (CPM) Analysis 15

6 Cited References **16**

Appendices **17**

A Updated Design Requirements **17**

B Concept Screening and Scoring Matrices **17**

C Engineering Drawing Package **17**

D Product Data Sheets and Price Quotes **17**

E Detailed Calculations **18**

List of Figures

1 Isometric view of the complete TCD assembly. Arrows indicate the actuator subsystem (left) and valve subsystem (right). Flow enters from the inlet port and exits through the outlet port. 7
2 Controls diagram of the TCD. Outlines the overall flow of inputs and outputs 7
3 Actuator sub-assembly. The numbered components are: (1) **[TODO: component]**, (2) **[TODO: component]**, (3) **[TODO: component]**. 8
4 Valve sub-assembly showing the flow restriction mechanism. 9
5 Updated Gantt chart spanning Winter and Spring quarters. The vertical dashed line indicates the current date. 15
6 Critical path method (CPM) network diagram. The critical path is highlighted in red. 16

List of Tables

1 Example requirements rolldown from system to subsystem level. 10
2 Preliminary Bill of Materials and Cost Estimate 13
3 Updated task list for Winter and Spring quarters. 14

1 Abstract

Sandia National Laboratories (SNL) has tasked our senior design team, Noble Aggie Labs to design a passive temperature compensation device (PTCD) that regulates flow in a pipe. Flow must be regulated by adjusting an orifice geometry according to a change in ambient air temperature. The PTCD must be purely thermo-mechanically driven, functioning without the reliance on embedded systems(electronics) or human input. The required mathematical relationship between the ambient temperature and a characteristic orifice geometry is provided by SNL. Additionally, materials compatibility with the fluid –primarily noble gases– must be considered. In order to meet these requirements, the PTCD will be divided into three main subsystems: (1) Actuator, (2) Interface, (3) Valve Mechanism.

This Preliminary Machine Architecture Report outlines how well the current iteration of the three main subsystems conform to the specified mathematical relationship. Explicitly, the relationship can be summarized as $D_{\text{eff}} = A_1(T_{\text{ambient}} - T_{\text{ref}}) + A_0$ which will be referred to as the "required relationship" throughout this report. The PTCD and its constituent subsystems must facilitate this linear relationship between the characteristic orifice diameter and the ambient temperature.

2 System Architecture Selected

In order to more easily The relationship between the ambient temperature and its corresponding orifice geometry is governed by the following equation:

$$D_{\text{eff}} = A_1(T - T_{\text{ref}}) + A_0 \tag{1}$$

Be advised, $A_1 = -1.524 \times 10^{-3} \pm 5\% \frac{\text{mm}}{°\text{C}}$, $A_0 = 0.254 \pm 5\%$ mm, and $T - \text{ref} = 25\,°\text{C}$. The device must operate between $-60\,°\text{C}$ and $80\,°\text{C}$, withstand pressure differentials up to 68.947 57 MPa during choked flow, fit within a $(10\,\text{cm})^3$ mechanical envelope, and be compatible with hydrogen gas.

All design concepts were then systematic screened and weighted with various scoring matrices (Report 1). We have since converged on a single architecture consisting of three primary subsystems: (1) a **aluminum thermal expansion rod** a singular metallic rod that responds to ambient temperature changes. The aluminum rod will have linear piston displacement as a result of thermal exapnsion via the following equation:

$$\Delta L = L_0 \, \Delta T \, \alpha \tag{2}$$

(2) A **compound spur-gear train** comprising of two rack gears and two coaxial pinion gears with a gear ratio of 4:1. This amplifies and transmits the piston's linear motion to the valve; and (3) a **sliding-plate orifice valve** ("pizza cutter" design), in which two flat plates, one fixed and one translating contain a shaped hole with the overlap between the two holes defining the effective flow orifice. As the actuator extends with increasing temperature, the moving plate translates downwards. This reduces the overlap area and shriking D_{eff} per the requirements.

2.1 Selection Criteria and Design Justification

[TODO: Include volume specification and volume of subsystems]

[TODO: State the single concept being carried forward. For example: "After evaluating five actuator concepts and ten valve concepts through systematic screening and weighted scoring matrices (see Appendix B), our team selected the [Actuator Type] coupled with a [Valve Type] as our final system architecture."]

[TODO: Discuss the selection criteria used:

- **Temperature range compatibility ($-60\,^\circ\text{C}$ to $80\,^\circ\text{C}$)**
- **Mechanical envelope ($10\,\text{cm}$ cube)**
- **Hydrogen compatibility of wetted materials**
- **Ability to withstand pressure differentials up to $10\,000\,\text{psid}$**
- **Linear or controllable relationship between temperature and flow restriction**
- **Simplicity and reliability (passive operation, no electronics)**

]

[TODO: Explain why each subsystem solution was chosen. Emphasize STRENGTHS, not weaknesses of alternatives.]

2.2 Complete System Overview

[TODO: Provide a 1–2 paragraph overview of how the entire system works at a high level. Walk the reader through the flow of energy/information: temperature

change → actuator displacement → valve position change → flow restriction adjustment.]

[Full System Assembly — Isometric View]

Include arrows pointing to each subsystem.
Label: Actuator, Valve, Housing, Inlet, Outlet.

Figure 1: *Isometric view of the complete TCD assembly. Arrows indicate the actuator subsystem (left) and valve subsystem (right). Flow enters from the inlet port and exits through the outlet port.*

[TODO: Reference the figure: "As shown in Figure 1, the system consists of. . ."]

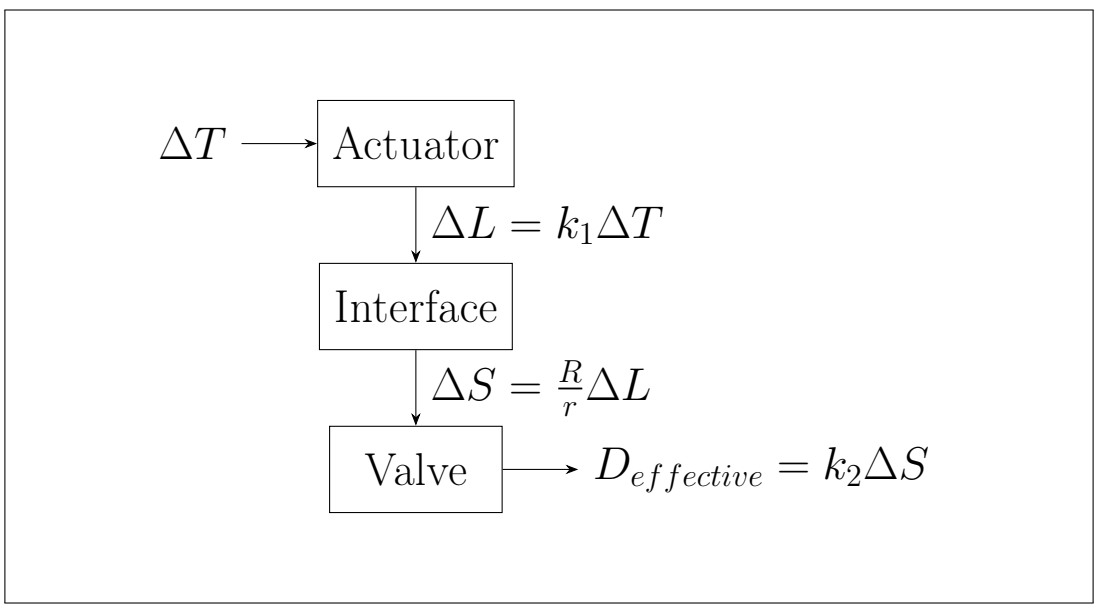

Figure 2: *Controls diagram of the TCD. Outlines the overall flow of inputs and outputs*

2.3 Subsystem Overviews

2.3.1 Thermal Actuator

[TODO: Describe the actuator subsystem. Include:

- **Operating principle (how temperature change creates displacement)**
- **Key components and their functions**
- **Why this actuator type was selected (reference scoring matrix)**
- **Expected displacement range over the temperature range**

]

[Actuator Sub-Assembly Drawing]

Exploded view with numbered callouts.
Show how displacement is generated.

Figure 3: *Actuator sub-assembly. The numbered components are: (1) **[TODO: component]**, (2) **[TODO: component]**, (3) **[TODO: component]**.*

2.3.2 Valve

[TODO: Describe the valve subsystem. Include:

- **Valve type and operating principle**
- **How actuator displacement translates to flow area change**
- **Key components**
- **Expected flow coefficient (Cv) range**

]

[Valve Sub-Assembly Drawing]

Show open and closed positions.
Use dashed lines for alternate positions.

Figure 4: *Valve sub-assembly showing the flow restriction mechanism.*

2.3.3 Housing and Fluid Interface

[TODO: Describe the housing that contains everything. Include:

- **Material selection (hydrogen compatibility, steel/aluminum)**
- **Sealing strategy (O-rings, gaskets, welding)**
- **How the device connects to the existing piping system**
- **Thermal conduction path from ambient air to the actuator**

]

3 Discussion

3.1 Design Requirement Updates

[TODO: Discuss how requirements changed since Report 1. For example:

- **Did discussions with Dr. Bradley Bon reveal new constraints?**
- **Did preliminary calculations show any requirements were infeasible or needed adjustment?**
- **Were any requirements added, removed, or modified?**

Reference the updated requirements table in Appendix A.]

3.2 Requirements Rolldown to Subsystems

[TODO: Explain how system-level requirements were decomposed into subsystem-level specifications. This is critical for the grading rubric.]

Table 1: *Example requirements rolldown from system to subsystem level.*

System Req.	Spec	Subsystem Rolldown	Sub-Spec
Temperature range	$-60\,^{\circ}\mathrm{C}$ to $80\,^{\circ}\mathrm{C}$	The actuator must produce monotonic displacement over this full range	TODO
Mechanical envelope	$\leq 10\,\mathrm{cm}^3$	TODO: Actuator $\leq$ X cm, Valve $\leq$ Y cm, Housing $\leq$ Z cm	TODO
Pressure	ΔP $\leq$ 68.94757 MPa	Valve must not deform or leak under max pressure	TODO
Effective diameter	$D_{\text{effective}}$ = 0.17018 mm - 0.38354 mm	TODO: Valve must not deform or leak under max pressure	TODO
Tolerance	$D_{\text{effective}} = \pm 0.0127$ mm	TODO: Valve must not deform or leak under max pressure	TODO
H_2 compatibility	All wetted parts	TODO: Specify material for each wetted component	TODO

3.3 Subsystem Interactions and Technical Detail

[TODO: This is where you go DEEP into the technical details. Unlike Section 2 which was a high-level overview, this section provides engineering proof that the design works.]

[TODO: Describe how subsystems interface with each other:

- **How does the actuator physically connect to the valve?**
- **What is the mechanical coupling (direct contact, linkage, etc.)?**
- **How does thermal energy flow through the system?**
- **What are the potential failure modes at interfaces?**

]

3.4 Design Verification Calculations

[TODO: Introduce the purpose of these calculations: to demonstrate analytically that the design can meet its requirements.]

3.4.1 Thermal Expansion and Actuator Displacement

[TODO: Show that the actuator produces sufficient displacement over the temperature range to fully open/close the valve.]

[TODO: Replace this example with your actual thermal expansion calculation.]

The linear thermal expansion of a rod is given by:

$$\Delta L = \alpha \cdot L_0 \cdot \Delta T \tag{3}$$

where:

- ΔL is the change in length [mm],
- α is the coefficient of thermal expansion [°C^{-1}],
- L_0 is the original length [mm],
- ΔT is the temperature change [°C].

  Assumptions:

1. **[TODO: Material properties are constant over the temperature range.]**
2. **[TODO: Thermal equilibrium is reached before the test window.]**
3. **[TODO: List all assumptions.]**

[TODO: Substitute your values into Equation 3 and show the result. Compare the result to the requirement.]

3.4.2 Force Analysis Under Pressure Loading

[TODO: Show that the actuator can generate enough force to move the valve against the pressure differential. Consider worst case: 10 000 psid.]

Paramter values:

- $E =$
- $A =$
- $\alpha =$
- $\Delta T =$

$$F = EA\alpha\Delta T \tag{4}$$

[TODO: Insert your actual force/pressure calculations here.]

3.4.3 Performance Relationship

[TODO: Demonstrate that the system achieves the required flow characteristic (e.g., the relationship between temperature and effective flow area or discharge coefficient). Reference Performance_Relationship.xlsx if applicable.]

3.4.4 Additional Analyses

[TODO: Add any other calculations that prove feasibility:

- **Heat transfer analysis (thermal response time)**
- **Stress/strain analysis of critical components**

- **Material compatibility verification**
- **Tolerance stack-up analysis**
- **FEA results (if performed)**

]

4 Preliminary Bill of Materials

[TODO: Write a brief introductory paragraph explaining the BOM structure and the total estimated project cost.]

Table 2: *Preliminary Bill of Materials and Cost Estimate*

Item	Part Name	Description / Supplier	Qty	Material	P/
Actuator Subsystem					
1	[TODO: Part name]	[TODO: Description / McMaster #]	[TODO: 1]	[TODO: Steel]	P
2	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: Al 6061]	F
Valve Subsystem					
3	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: SS 316]	P
4	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: Brass]	F
Housing & Interface					
5	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: SS 316]	F
6	[TODO: Part name]	[TODO: Description]	[TODO: 2]	[TODO: Viton]	P
Fasteners & Miscellaneous					
7	[TODO: Part name]	[TODO: Description]	[TODO: 10]	[TODO: Steel]	P
Special Tooling					

(continued from previous page)

Item	Part Name	Description / Supplier	Qty	Material	P/
8	[TODO: Tooling item]	[TODO: Description]	[TODO: 1]	—	P
					Estimated Tota

Key: P = Purchased, F = Fabricated

[TODO: Add any notes about budget constraints, contingency funds, or items awaiting price quotes.]

5 Project Management Information

5.1 Updated Task List

[TODO: Provide a detailed task list. Organize by phase or subsystem. Must include:

- **Technical tasks (design, fabrication, testing)**
- **Deliverables for class (reports, presentations)**
- **Team and stakeholder meetings**
- **Procurement / ordering deadlines**

]

Table 3: *Updated task list for Winter and Spring quarters.*

#	Task Description	Owner(s)	Deadline
1	Finalize engineering drawings	[TODO: Name(s)]	[TODO: Date]
2	Order raw materials	[TODO: Name(s)]	[TODO: Date]
3	Machine actuator housing	[TODO: Name(s)]	[TODO: Date]
4	Assemble prototype	[TODO: Name(s)]	[TODO: Date]
5	Conduct thermal testing	[TODO: Name(s)]	[TODO: Date]
6	Conduct pressure testing	[TODO: Name(s)]	[TODO: Date]
7	[TODO: Add more tasks]		

5.2 Gantt Chart

[TODO: Insert your Gantt chart. This is typically created in Excel, MS Project, or a similar tool and exported as an image.]

[Gantt Chart — Two Quarters]

Export from Excel/MS Project as PNG or PDF.
Must span Winter 2026 through Spring 2026.
Show current progress with a vertical "today" line.

Figure 5: *Updated Gantt chart spanning Winter and Spring quarters. The vertical dashed line indicates the current date.*

[TODO: Write a paragraph interpreting the Gantt chart. Highlight key milestones and any tasks that are ahead of or behind schedule.]

5.3 Critical Path Method (CPM) Analysis

[TODO: Insert your CPM network diagram. This must be in GRAPHICAL format.]

[CPM Network Diagram]

Show tasks as nodes, dependencies as arrows.
Highlight the critical path in red or bold.
Include earliest start/finish and latest start/finish times.

Figure 6: *Critical path method (CPM) network diagram. The critical path is highlighted in red.*

[TODO: Write a paragraph describing:

- **What the critical path is (list the tasks)**
- **Total project duration along the critical path**
- **Which tasks have float (slack time)**
- **What the analysis tells you about risk and scheduling**
- **Any mitigation strategies for critical-path delays**

]

6 Cited References

References

[1] B. Bon, "Passive Temperature Compensation Device — Project Prompt," Sandia National Laboratories, SAND2025-15446O, 2025.

[2] **[TODO: Add reference: textbook, paper, or technical standard]**

[3] **[TODO: Add reference: material property data source]**

[4] **[TODO: Add reference: relevant prior work or patent]**

Appendices

A Updated Design Requirements

[TODO: Include the final, updated requirements table here. This should reflect any changes discussed in Section 3.1. Organize by category (system-level, actuator, valve, housing, etc.). Include the subsystem-level rolldown specifications.]

B Concept Screening and Scoring Matrices

[TODO: Include the screening matrix and weighted scoring matrix from your concept selection process. These were referenced in Section 2.1.]

C Engineering Drawing Package

[TODO: Include complete assembly, sub-assembly, and part drawings. Part drawings must include dimensions and tolerances for all parts that will be MANUFACTURED (not purchased). Follow proper engineering drawing practices.

Organize as:

1. **Full system assembly drawing**
2. **Actuator sub-assembly drawing**
3. **Valve sub-assembly drawing**
4. **Housing sub-assembly drawing**
5. **Individual part drawings (manufactured parts only)**

Each drawing should be on its own page for clarity. Label as Figure A.1, A.2, etc.]

D Product Data Sheets and Price Quotes

[TODO: Include manufacturer spec sheets for all purchased components. Include price quotes or screenshots of supplier pricing. This supports the Bill of Materials in Section 4.]

E Detailed Calculations

[TODO: If your calculations in Section 3.4 were summarized for brevity, include the full derivations and spreadsheet screenshots here. Reference Performance_Relationship.xlsx and Thermal_Expansion.xlsx data as appropriate.]